combined_locus_final.m

User Guide & Data Derivation Reference

Step 3 of the IPV Benchmarking Pipeline · SEE 894 · Simon Fraser University · Spring 2026

Purpose

This is the final step of the IPV benchmarking pipeline. It takes the locus coordinates produced by spectral_projection_locus.m for each of the six approved devices, plots all loci together on a single figure, and performs a dominance analysis. The result is the combined PCE vs λ_onset,95% comparison — the primary figure in the project report showing the three-tier performance hierarchy across perovskite, OPV, and inorganic selenium technologies.

1. How This Script Fits Into the Pipeline

Step	What It Does
Step 1 — ipv_validation.m	Validates each paper's digitised data. Produces Jsc, Voc, FF, PCE, and λ_onset,95%. Papers that pass all three checks are APPROVED.
Step 2 — spectral_projection_locus.m	Projects each approved device's J–V curve onto the four corner LEDs. Produces four (λ_onset,95%, PCE) coordinates — the corners of the device's locus.
Step 3 — combined_locus_final.m (this script)	No file I/O. All locus coordinates are hardcoded directly from Step 2 results. Plots all device loci together, produces a console summary table, and performs pairwise dominance analysis.

No CSV files needed

Unlike Steps 1 and 2, this script requires no CSV input files. All device data — the four PCE corner values and λ_onset,95% coordinates — are hardcoded in the DEVICE DATA section at the top of the script. You populate these by copying the LOCUS COORDINATES output from Step 2 for each device.

2. Understanding the Data Structure

Each device is represented by a MATLAB struct with five fields. Here is the complete entry for Device A:

```
devices(1).label = 'Device A (PVSK, 1.67 eV)';  % display name in legend
devices(1).tech  = 'Perovskite';                 % technology class (for console
table)
devices(1).Eg    = 1.67;                         % absorber bandgap in eV
devices(1).l95   = [645.8, 660.1, 702.0, 721.1]; % λ_onset,95% for Ref4,3,2,1 (nm)
devices(1).PCE   = [40.58, 43.16, 41.18, 43.76]; % PCE at Ref4,3,2,1 (%)
devices(1).color = [0.12 0.47 0.71];             % RGB colour for this device's
locus
```

Corner order convention

The four values in I95 and PCE always follow the order: [Ref4, Ref3, Ref2, Ref1] = [644 nm, 660 nm, 702 nm, 721 nm]. This order matches the column order in LED_spectra.csv and the output order from spectral_projection_locus.m. Do not reorder.

3. The Six Devices in This Script

The following devices are included in the current version of the script:

Device	Paper / Author	Tech	Eg (eV)	Locus Colour
Device A	S3 Chen 2025	PVSK	1.67	Blue [0.12 0.47 0.71]
Device B	S4 Wen 2025	PVSK	1.79	Teal [0.00 0.62 0.45]
Device C	S5 Liu 2024	PVSK	1.56	Pink [0.80 0.47 0.65]
Device D	S8 Li 2024	PVSK	1.52	Purple [0.58 0.40 0.74]
Device E	S12 Wang 2024	OPV	1.74	Orange [0.89 0.47 0.20]
Device F	S15 Lu 2024	Inorganic Se	1.90	Red [0.84 0.15 0.16]

4. The Four Corner Reference Points

The four x-axis positions are fixed across all devices — they are properties of the corner LEDs, not of the devices:

Reference / Position	Description
Ref 4 | 644.8 nm	5700K standard phosphor LED. Most blue-dominant, lowest red content. Plotted in red. Leftmost x-position.
Ref 3 | 660.1 nm	3000K full-spectrum (FS) LED. Plotted in orange.
Ref 2 | 702.0 nm	5700K full-spectrum (FS) LED. High red content despite cool colour temperature. Plotted in blue.
Ref 1 | 721.1 nm	3000K KSF phosphor LED. Most red-rich of the four. Plotted in green. Rightmost x-position.

Note: In the script, `I95` is set to these fixed values for every device. The λ_onset,95% of the corner LEDs does not change between papers — only the PCE values change.

5. How to Add a New Device

To add a new approved device to the combined plot:

1. **Run Step 1 (ipv_validation.m)** for the new paper. Confirm all three checks PASS.

2. **Run Step 2 (spectral_projection_locus.m)** for the paper. Copy the LOCUS COORDINATES block from the console output.
3. **Open combined_locus_final.m** and find the DEVICE DATA section.
4. **Add a new struct entry** at the end of the list. Increment the index number:

```
devices(7).label = 'Device G (PVSK, 1.71 eV)';
devices(7).tech  = 'Perovskite';
devices(7).Eg    = 1.71;
devices(7).l95   = [645.8, 660.1, 702.0, 721.1];   % always these fixed values
devices(7).PCE   = [XX.XX, XX.XX, XX.XX, XX.XX];   % from Step 2 LOCUS COORDINATES
devices(7).color = [0.12 0.47 0.71];               % choose a colour for this tech
```

5. **Set locus_color** from the technology colour guide (same as in spectral_projection_locus.m).
6. **Run the script.** The new device will appear automatically on the figure and in the console summary and dominance analysis.

> **l95 values are always fixed**
> Every device always uses l95 = [645.8, 660.1, 702.0, 721.1] nm. These are the λ_onset,95% values of the four corner LEDs. They do not depend on the paper. Do not copy the source l95 value from Step 2 into this field — only copy the four corner PCE values.

6. How to Remove or Exclude a Device

To temporarily exclude a device from the plot without deleting its data, comment out its struct block:

```
% devices(3).label = 'Device C ...';  % commented out — excluded from plot
% devices(3).tech  = 'Perovskite';
% devices(3).Eg    = 1.56;
% ...
```

MATLAB will automatically re-number the remaining struct entries. The plot, console table, and dominance analysis will only include the uncommented devices.

7. How to Change a Device's Colour

Each device has an RGB colour triplet [R G B] where each value is between 0 and 1. The colour is used for the locus fill, the locus outline edges, and any associated text labels.

Technology	Recommended colour [R G B]
Perovskite (PVSK)	[0.12 0.47 0.71] blue (primary)
Perovskite variant 2	[0.00 0.62 0.45] teal
Perovskite variant 3	[0.80 0.47 0.65] pink
Perovskite variant 4	[0.58 0.40 0.74] purple

Organic PV (OPV)	[0.89 0.47 0.20] orange
DSSC	[0.17 0.63 0.17] green
a-Si:H	[0.58 0.40 0.74] purple
Inorganic Se	[0.84 0.15 0.16] red

Tip
If you have multiple devices of the same technology, use different colours within the same hue family (e.g. dark blue, medium blue, light blue for three perovskite devices) so they are visually distinguishable while still appearing related.

8. The Locus Quadrilateral

8.1 What the Quadrilateral Represents

Each device's locus is a filled quadrilateral in PCE–λ space. Its four corners are the (λ_onset,95%, PCE) coordinates at the four corner LEDs. The interior of the quadrilateral represents the complete range of PCE values the device would achieve under any realistic indoor LED — not just the four corners.

8.2 How the Quadrilateral Is Drawn

The quadrilateral vertices are connected in the order: Ref4 → Ref2 → Ref1 → Ref3 → back to Ref4. This ordering (`quad_idx = [1,3,4,2,1]`) places the two 5700K points on one diagonal and the two 3000K points on the other, forming a proper parallelogram shape:

```
quad_idx = [1, 3, 4, 2, 1];   % index order: Ref4 -> Ref2 -> Ref1 -> Ref3 -> Ref4

% This connects:
%   Ref4 (644nm, 5700K std)  ->  Ref2 (702nm, 5700K FS)   [bottom 5700K edge]
%   Ref2 (702nm, 5700K FS)   ->  Ref1 (721nm, 3000K KSF)  [right edge]
%   Ref1 (721nm, 3000K KSF)  ->  Ref3 (660nm, 3000K FS)   [top 3000K edge]
%   Ref3 (660nm, 3000K FS)   ->  Ref4 (644nm, 5700K std)  [left edge]
```

The shape is filled at 18% opacity (`FaceAlpha = 0.18`) so that overlapping loci remain readable. The outline is drawn at full opacity.

8.3 The Region Labels α β γ δ

Four Greek letters label the regions between the vertical dashed lines at each corner λ_onset,95% value:

Region	λ Range and Meaning

α (alpha)	< 644 nm — to the left of Ref 4. Outside the standard indoor range. Devices measured here (like Device F at 616 nm) are characterised under non-representative light.
β (beta)	644–660 nm — between Ref 4 and Ref 3. High CCT, moderate spectral reach.
γ (gamma)	660–702 nm — between Ref 3 and Ref 2. Broad middle range.
δ (delta)	≥ 702 nm — to the right of Ref 2 (and Ref 1 at 720 nm). High red content, most favourable for wider-bandgap absorbers.

8.4 Corner Dot Colours

Each corner dot on the locus plot is coloured by the reference LED identity, not by the device:

Corner Dot Colour	Reference LED
Red dot	Ref 4 — 5700K std (644 nm)
Orange dot	Ref 3 — 3000K FS (660 nm)
Blue dot	Ref 2 — 5700K FS (702 nm)
Green dot	Ref 1 — 3000K KSF (721 nm)

9. Console Output: Summary Table

The script prints a summary table of all devices followed by a dominance analysis. Here is the full expected output for the six-device configuration:

```
==================================================================================
 COMBINED LOCUS SUMMARY
==================================================================================
Device                          Technology    Eg    Ref4    Ref3    Ref2    Ref1    PCE
range
----------------------------------------------------------------------------------
Device A (PVSK, 1.67 eV)        Perovskite  1.67  40.58   43.16   41.18   43.76
40.58-43.76%
Device B (PVSK, 1.79 eV)        Perovskite  1.79  39.66   41.58   37.89   38.16
37.89-41.58%
Device C (PVSK, 1.56 eV)        Perovskite  1.56  37.22   40.20   38.30   41.59
37.22-41.59%
Device D (PVSK, 1.52 eV)        Perovskite  1.52  39.24   41.46   40.28   42.85
39.24-42.85%
Device E (OPV, 1.74 eV)         OPV         1.74  28.41   30.50   27.97   29.34
27.97-30.50%
Device F (Se, 1.90 eV)          Inorganic   1.90  16.64   14.97   14.18   11.08
11.08-16.64%
```

Column	Meaning
Device	Display label from devices(i).label.
Technology	Technology class from devices(i).tech.

Eg	Absorber bandgap in eV from devices(i).Eg.
Ref4 / Ref3 / Ref2 / Ref1	PCE (%) at each of the four corner conditions.
PCE range	min(PCE) – max(PCE) across all four corners.

10. Console Output: Dominance Analysis

After the summary table, the script performs a pairwise dominance check between every pair of devices:

```
---------------------------------------------------------------------------------
 DOMINANCE ANALYSIS
---------------------------------------------------------------------------------
  Device A (PVSK, 1.67 eV) DOMINATES Device B (PVSK, 1.79 eV) at ALL 4 corners

  Device A (PVSK, 1.67 eV) DOMINATES Device C (PVSK, 1.56 eV) at ALL 4 corners

  Device A vs Device D
    Former wins at: Ref4(644nm), Ref3(660nm), Ref2(702nm), Ref1(720nm)
    ...

  Device A (PVSK, 1.67 eV) DOMINATES Device E (OPV, 1.74 eV) at ALL 4 corners

  Device A (PVSK, 1.67 eV) DOMINATES Device F (Se, 1.90 eV) at ALL 4 corners
=================================================================================
```

Output type	Meaning
X DOMINATES Y at ALL 4 corners	Device X has higher PCE than Device Y at every single corner condition. The loci do not overlap — X is unconditionally superior to Y across the full realistic indoor LED range.
X vs Y / Former wins at: ... / Latter wins at: ...	Neither device dominates the other completely. Each is superior at some corner conditions. The specific corners where each wins are listed.

Interpreting dominance

A dominance result means the device is better across ALL realistic indoor LEDs — not just on average. If a device dominates another, no choice of LED can make the dominated device perform better. Partial wins (crossover cases) indicate spectral sensitivity: the relative ranking depends on which LED is used.

11. The Combined Locus Data (Current Six Devices)

The following table shows all PCE corner values hardcoded in the current script, as derived from spectral_projection_locus.m in Step 2:

Device	Eg (eV)	PCE Ref4 644.8 nm	PCE Ref3 660.1 nm	PCE Ref2 702.0 nm	PCE Ref1 721.1 nm

A — PVSK 1.67 eV	1.67	40.58	43.16	41.18	43.76
B — PVSK 1.79 eV	1.79	39.66	41.58	37.89	38.16
C — PVSK 1.56 eV	1.56	37.22	40.20	38.30	41.59
D — PVSK 1.52 eV	1.52	39.24	41.46	40.28	42.85
E — OPV 1.74 eV	1.74	28.41	30.50	27.97	29.34
F — Se 1.90 eV	1.90	16.64	14.97	14.18	11.08

Device F note

Device F (Se, 1.90 eV) was characterised under a source LED with λ_onset,95% = 616.8 nm — outside the realistic indoor range. Its PCE values decline from Ref4 to Ref1 (opposite trend to all other devices) because its bandgap of 1.90 eV is too wide for the longer-wavelength corner LEDs. The reported source PCE of 18.25% significantly overestimates its realistic indoor performance.

12. Step-by-Step Usage

7. **Complete Steps 1 and 2** for every device you want to include. All must have passed validation.
8. **Open combined_locus_final.m.** Find the DEVICE DATA section at the top.
9. **For each device, verify** that the four PCE values in `devices(i).PCE` match the LOCUS COORDINATES output from Step 2.
10. **To add a new device:** append a new struct block with the correct index, copy the four PCE corner values from Step 2, use the fixed I95 = [645.8, 660.1, 702.0, 721.1], and set the colour.
11. **Run the script** (press F5 or Run). MATLAB produces the figure and prints the console summary.
12. **Check the figure:** all device loci should appear. Loci that do not overlap with others represent dominated performance bands.
13. **Read the dominance analysis** in the console to identify which devices unconditionally outperform others.
14. **Export the figure:** right-click the figure → Save As → PNG or PDF for inclusion in the report.

13. Troubleshooting

Symptom	Cause and Fix
A device's locus appears as a horizontal line (not a quadrilateral)	All four PCE values are identical or nearly identical. This is physically correct for very flat loci (e.g. a device whose EQE spans the full visible range). It is not an error.
Loci overlap excessively and are hard to distinguish	Increase FaceAlpha (e.g. from 0.18 to 0.10) to make fills more transparent. Or change the colours to more distinct hues.
Dominance analysis shows no full dominances	This is physically possible if all devices have at least one corner where rankings flip. Review the PCE values to confirm they are entered correctly in Ref4,Ref3,Ref2,Ref1 order.

PCE values look wrong compared to Step 2 output	Check that you copied the corner PCE values (from the LOCUS COORDINATES table), not the source PCE or the Step 1 PCE_calc.
One device's locus has the opposite slope to all others (PCE decreasing with λ)	This indicates the device's bandgap is too wide for longer-wavelength LEDs. Device F (Se, 1.90 eV) shows this naturally. If unexpected, verify the PCE column order: [Ref4, Ref3, Ref2, Ref1].
Figure legend is too large / overlapping the plot	Change legend position: replace 'northoutside' with 'southwest' or another location, or reduce FontSize in the legend call.

14. Reference

The locus-based benchmarking framework is from:

Khampa et al. (2026). *"A locus-based benchmarking framework for indoor photovoltaics."* Newton, 2, 100437.

Project report: *"A Universal Design and Benchmarking Framework for Indoor Photovoltaics."* Olarewaju Okikioluwa Victor & Ali Ghahremani. SEE 894, Simon Fraser University, Spring 2026.

This guide was generated from the source code of combined_locus_final.m. All data values, plot parameters, and analysis logic reflect the actual implementation.